rohitghosh2134 / **ChessEngine**

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rohitghosh2134 Final Changes e72ee96 · 5 months ago

images	Added Pictures	5 months ago
ChessBot.py	Initial commit: Added Chess Engine	5 months ago
ChessEngine.py	Initial commit: Added Chess Engine	5 months ago
ChessEngineAd.py	Initial commit: Added Chess Engine	5 months ago
ChessMain.py	Final Changes	5 months ago
LICENSE	Initial commit: Added Chess Engine	5 months ago
README.md	Initial commit: Added Chess Engine	5 months ago
config.py	Initial commit: Added Chess Engine	5 months ago
dockerfile	Made Changes to Dockerfile	5 months ago

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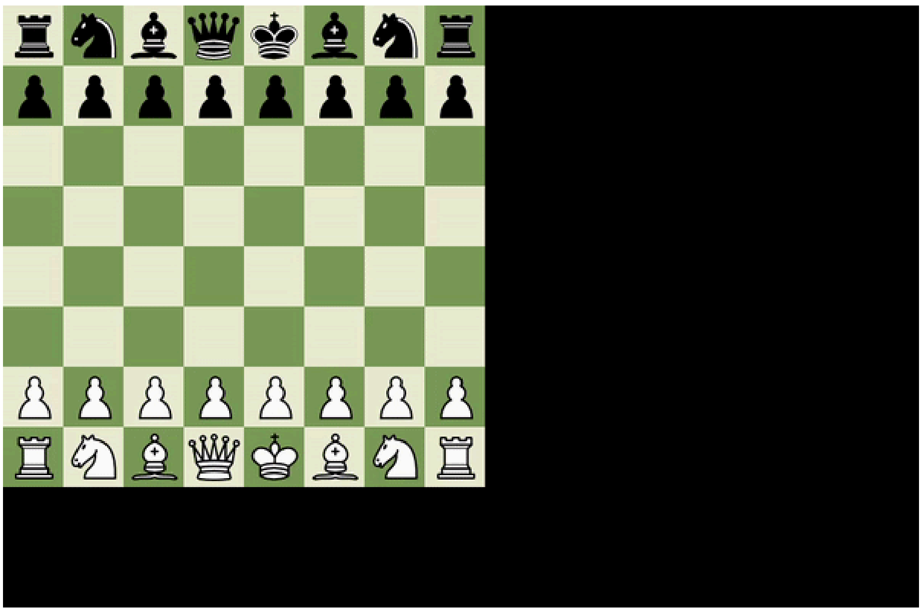
Chess



Files:

1. `ChessMain.py` : Handles user input and game visuals.
2. `ChessEngine.py` : Implements the chess logic using a brute-force algorithm.
3. `ChessEngineAd.py` : Implements an optimized chess algorithm.
4. `ChessBot.py` : Contains logic for a bot that evaluates captures, defenses, and positional advantages.
5. `config.py` : Configuration file to set player modes, board colors, and other parameters.

Dependencies:

- `pygame` : Install using `pip install pygame`

Running the Game

Running Locally (Without Docker)

To play the game using the optimized algorithm:

```
python ChessMain.py adv
```

To play using the brute-force algorithm:

```
python ChessMain.py
```

Running with Docker

1. Build the Docker Image

Run the following command in the project directory:

```
docker build -t chess-engine .
```

2. Run the Game

For brute-force algorithm:

```
docker run --rm -e DISPLAY=$DISPLAY -v /tmp/.X11-unix:/tmp/.X11-unix chess-engine
```

For optimized algorithm:

```
docker run --rm -e DISPLAY=$DISPLAY -v /tmp/.X11-unix:/tmp/.X11-unix chess-engine python ChessMain.py adv
```

3. Running with GUI on Linux:

- Ensure X11 forwarding is enabled by running:

  ```
  xhost +local:
  ```

- Then, start the container with:

  ```
  docker run --rm -e DISPLAY=$DISPLAY -v /tmp/.X11-unix:/tmp/.X11-unix chess-engine
  ```

4. Running with GUI on Windows (WSL2)

- Install an X server like **VcXsrv** or **Xming**.
- Set the `DISPLAY` variable:

```
export DISPLAY=$(ip route | awk '/default/ {print $3}'):0.0
```

- Then, run:

```
docker run --rm -e DISPLAY=$DISPLAY -v /tmp/.X11-unix:/tmp/.X11-unix chess-engine
```

Changing Player Modes in `config.py`

Modify `config.py` to choose who plays:

Mode	`PLAYER_ONE_HUMAN`	`PLAYER_TWO_HUMAN`
Play as White	`True`	`False`
Play as Black	`False`	`True`
2 Player Game	`True`	`True`
Watch AI Play	`False`	`False`

Now, your chess engine can run inside a Docker container while supporting GUI output! 🚀

Releases

No releases published
Create a new release

Packages

No packages published
Publish your first package

Languages

- **Python** 100.0%

Suggested workflows

Based on your tech stack

Pylint Configure

Lint a Python application with pylint.

Python application Configure

Create and test a Python application.

Python Package using Anaconda Configure

Create and test a Python package on multiple Python versions using Anaconda for package management.

More workflows

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